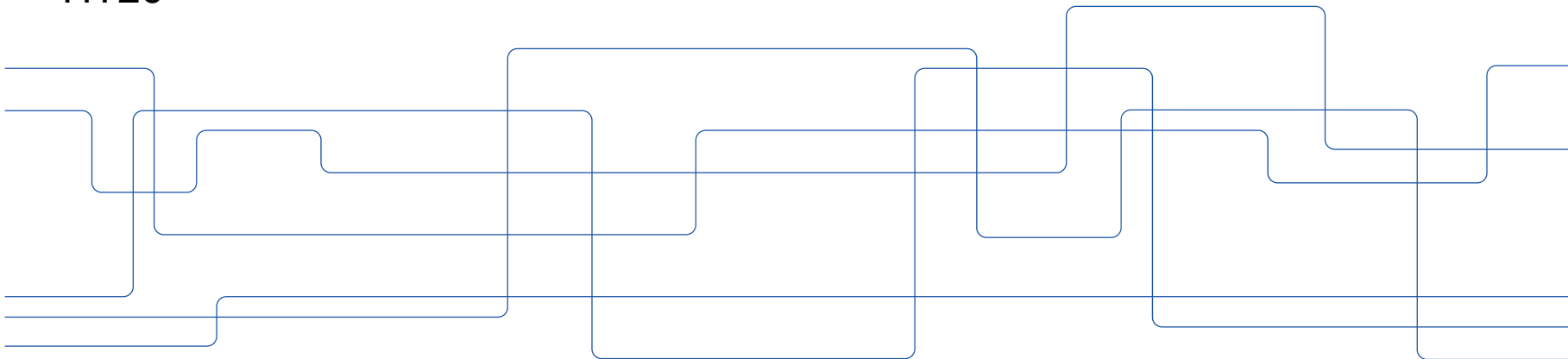


AI applications in Sustainable Energy Engineering Projects explanations and Group forming

HT26





Outlook

Group forming

Matching groups and projects

Projects explanation



Groups

5-6 students in each group

31 groups in total

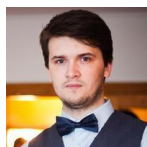
6 supervisors (each supervisor will supervise 5 groups)



Jagruti



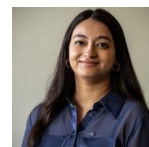
Marco



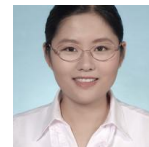
Taras



Farzin



Supriya



Yang



How to be assigned to a project

- I **randomly** assigned groups to different supervisors
- Topics are different, but methods are similar
- If you don't get your favorite topic, you still can follow other topics results

► Project groups 1-Marco 1 Project groups	5 students	🔒
► Project groups 2-Taras 1 Project groups	5 students	🔒
► Project groups 3-Jagruti 1 Project groups	6 students	🔒
► Project groups 4-Farzin 1 Project groups	5 students	🔒
► Project groups 5-Marco 2 Project groups	5 students	🔒
► Project groups 6-Taras 2 Project groups	5 students	🔒
► Project groups 7-Farzin 2 Project groups	5 students	🔒

Forecasting Day ahead electricity Market Prices Using Deep Learning Techniques

Jagruti Thakur

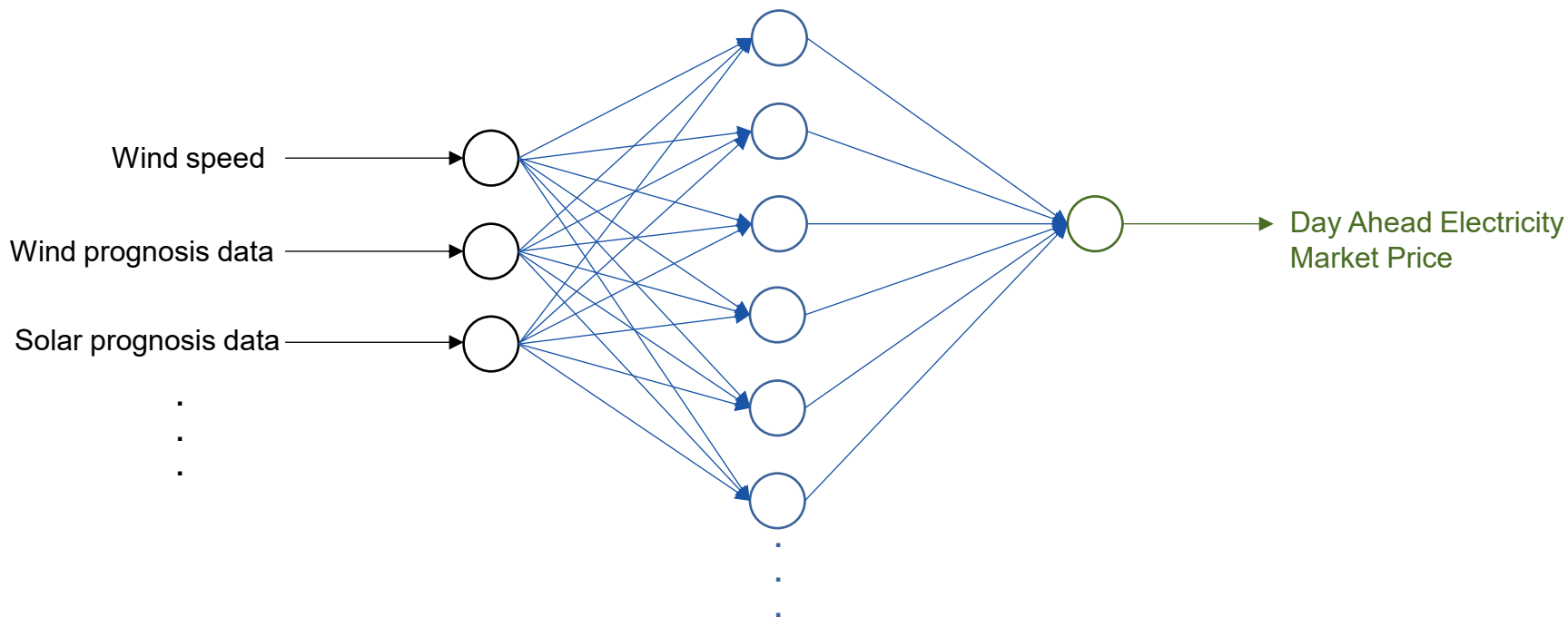
Digital Futures Faculty

jrthakur@kth.se





Forecasting Day ahead electricity Market Prices Using Deep Learning Techniques





Steps

- 1) Each group selects one EU country considering the fact that market data is available.
 - 2) **Step1-** Do a detailed literature review, to assess the work done, methods used, and results obtained. There are different ways other than AI to do forecasting
 - 3) **Step2-** Based on the initial study and review conducted, identify the variables needed for forecasting. You can use 1-5 years of historical data and forecast 1 week. Use hourly data for all the variables.
 - 4) **Step 3-** Renewable energy data can be used from renewables.ninja . For electricity market data use ENTSOE platform (<https://transparency.entsoe.eu/>) or you can use rebase.energy (instruction pdf given)
 - 5) **Step 4-** Check the data for missing values and replace them using certain logic. Explain the rationale behind it.
 - 6) **Step 5-** Setting up the model- Pre-processing step: Explain clearly what type of data manipulation techniques were used if any and why.
 - 7) **Step 6-** Design atleast two different scenarios based on government policies or target.
 - 8) **Step 7-** Forecast market prices using:
 - Logistic Regression
 - Shallow Neural Networks (1 hidden layer)
 - Deep Neural Networks (more than 1 hidden layer)
 - 5) **Step 8-** For each method, hyper tune the parameters to find the best performing parameters.
 - 6) **Step 9-** Compare different methods and report the best performing model
 - 7) **Step 10-** Compare your results with existing literature and discuss
-

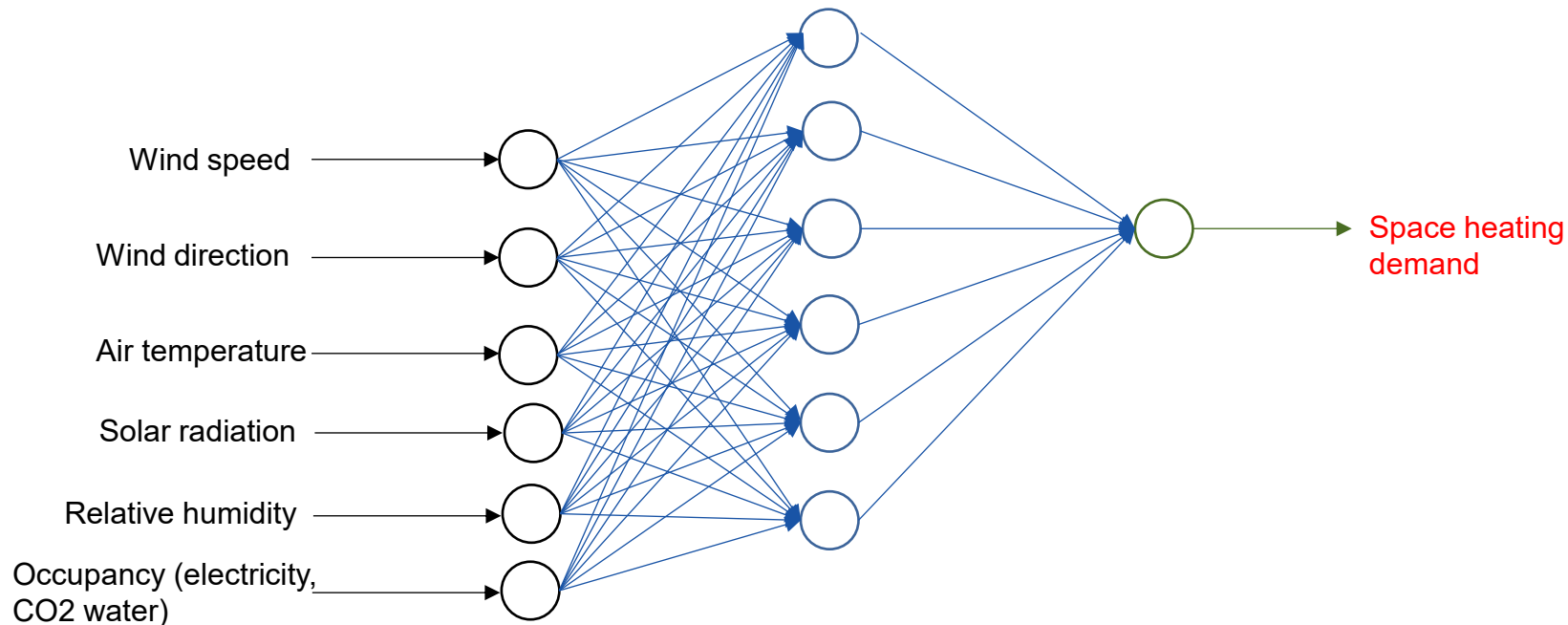
Forecasting Space Heating and Cooling Use Using Deep Learning Techniques: The UndervisningHus Case Study

Marco Molinari

marcomo@kth.se



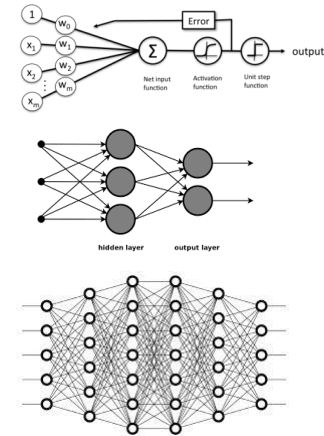
Title: Forecasting **Space Heating Use** Using Deep Learning Techniques: The Undervisninghus Case Study



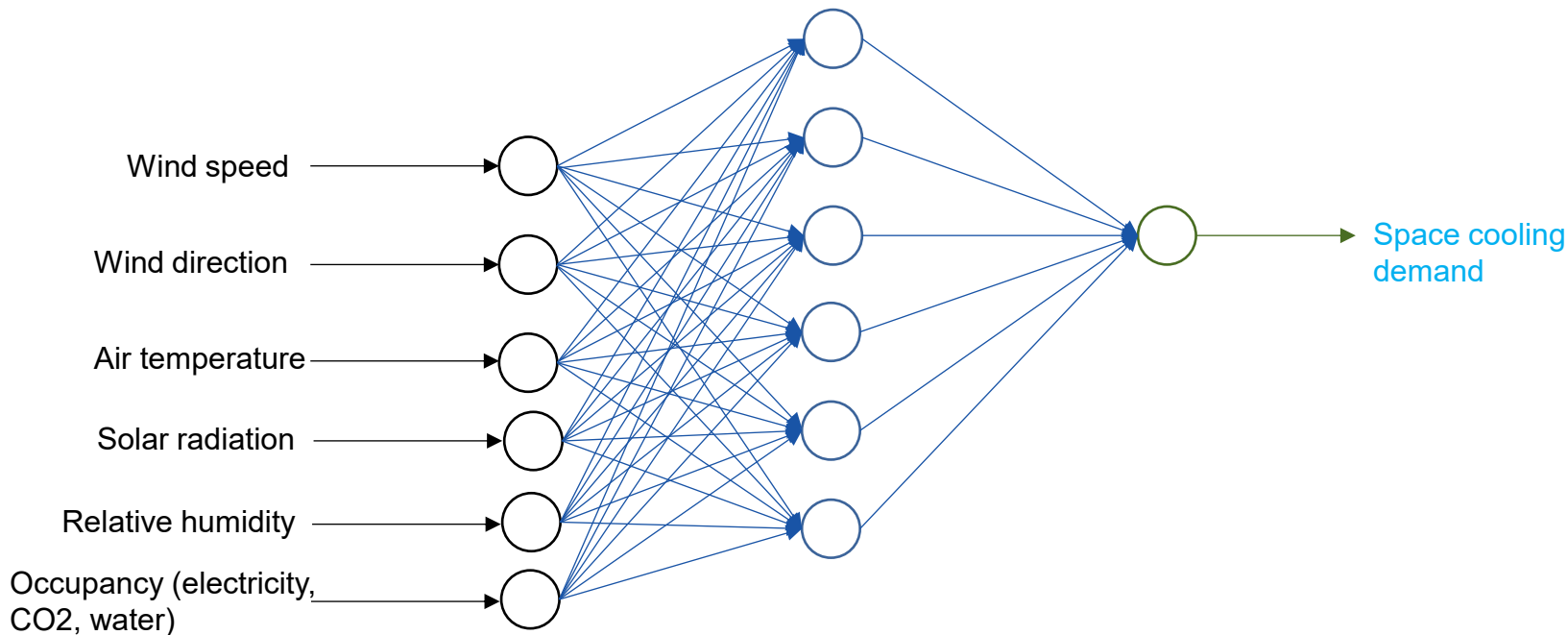
Title: Forecasting Building **Space Heating Use** Using Deep Learning Techniques: The Undervisninghus Case Study

Process:

- 1) Literature review
- 2) Collect historical data from the provided datasets
- 3) Clean and prepare the data
- 4) 1-day ahead Forecast energy used for heating using:
 - Logistic Regression
 - Shallow Neural Networks (1 hidden layer)
 - Deep Neural Networks (more than 1 hidden layer)
- 5) For each method, hyper tune the parameters to find the best performing parameters, and cross validation.
- 6) Compare different methods and report the optimum one
- 7) How would you address concerns related to the potential biases or inaccuracies in AI-based energy forecasts, especially when they influence critical decisions in sustainable energy systems?



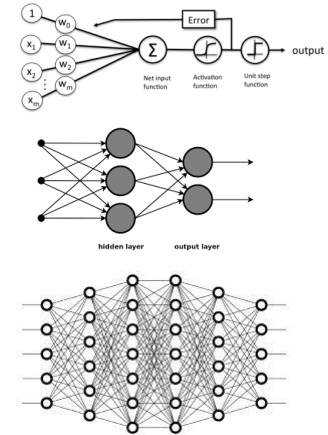
Title: Forecasting **Space Cooling Use** Using Deep Learning Techniques: The Undervisninghus Case Study



Title: Forecasting Building **Space Cooling Use** Using Deep Learning Techniques: The Undervisninghus Case Study

Process:

- 1) Literature review
- 2) Collect historical data from the provided datasets
- 3) Clean and prepare the data
- 4) 1-day ahead Forecast energy used for heating using:
 - Logistic Regression
 - Shallow Neural Networks (1 hidden layer)
 - Deep Neural Networks (more than 1 hidden layer)
- 5) For each method, hyper tune the parameters to find the best performing parameters, and cross validation.
- 6) Compare different methods and report the optimum one
- 7) How would you address concerns related to the potential biases or inaccuracies in AI-based energy forecasts, especially when they influence critical decisions in sustainable energy systems?



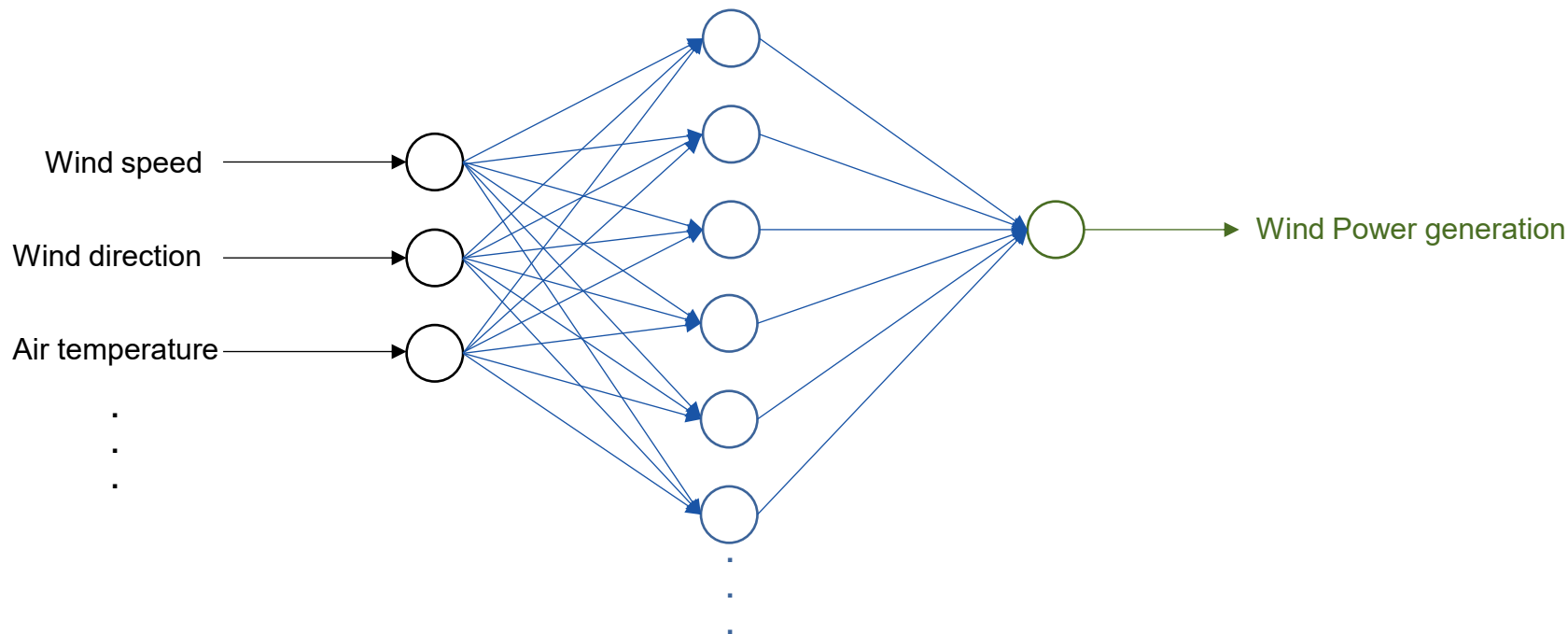
Wind Power Forecasting & Grid Integration Forecasting Using Machine Learning Techniques: A Case Study from Sweden (Luleå, Sundsvall, Stockholm, Malmö)

Taras Koturbash

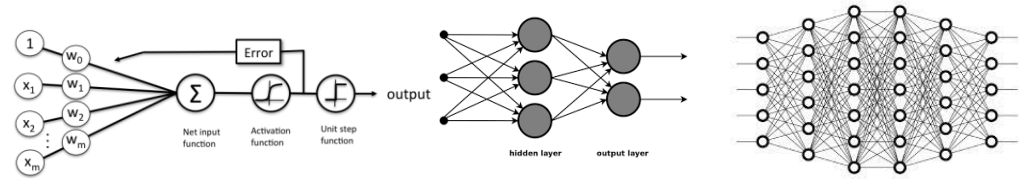
koturbas@kth.se



Title: Forecasting Wind Power Generation Using Deep Learning Techniques: A Case Study from Sweden



Projects: HPT



Title: Wind Power Forecasting Using Machine Learning: A Swedish Case Study (Luleå, Sundsvall, Stockholm or Malmö)

Investigate how input features, forecast horizon and model complexity affect prediction accuracy. Interpret your findings in the context of sustainable energy engineering, including the usefulness and limitations of forecasts for electricity-system operation.

- 1. Review ≥5 scientific papers:** identify input features, algorithms, and forecast horizons. Justify your choices.
- 2. Collect and prepare data:** use Renewables.ninja and/or SMHI; aim for ≥3 years of training data plus validation/test periods. Distinguish simulated from measured data.
- 3. Develop three models:** multiple linear regression, shallow neural network (1 hidden layer) and deep neural network (>1 hidden layer). Tune their hyperparameters.
- 4. Evaluate fairly:** use chronological training/validation/test splits and only inputs available when issuing the forecast. Compare MAE/RMSE against a persistence baseline.
- 5. Interpret the results:** discuss model complexity, forecast horizon, data limitations, and implications for energy-system operation.

Deliverables: concise report, reproducible code, and short presentation.



Wind Power Forecasting Using Machine Learning: A Swedish Case Study

Detailed Project Guidelines

Project objectives

- Develop and evaluate machine-learning models to forecast wind-power generation at a selected location in Sweden: Luleå, Sundsvall, Stockholm, or Malmö.
- Investigate how input features, forecast horizon, and model complexity affect prediction accuracy. Interpret your findings in the context of sustainable energy engineering, including the usefulness and limitations of forecasts for electricity-system operation.

1. Literature review and problem definition

- Review **at least 5** relevant scientific papers. Summarise their input data, forecasting methods, prediction horizons and evaluation approaches.
- Based on your review, formulate **a clear research question** and **justify your choice of input features**. Consider questions such as:
 - Does a deeper neural network improve forecasting accuracy compared with simpler models?
 - How does accuracy change with forecast horizon?
 - Which input features provide the greatest improvement?
- Use a common **one-hour-ahead forecast** as the main task. Investigate **one additional horizon**, such as six or 24 hours ahead.



Wind Power Forecasting Using Machine Learning: A Swedish Case Study

Detailed Project Guidelines

2. Data collection and preparation

- Collect **historical wind-power and meteorological data** for your selected location using Renewables.ninja and, where appropriate, SMHI. Aim to include at least 3 years for training, with additional periods reserved for validation and testing.
- **Document the data sources**, temporal resolution, units and relevant turbine assumptions. Clearly identify whether each dataset contains measurements or simulated values.
- **Inspect** missing values, anomalous observations and timestamp alignment. Explain and justify your preprocessing choices.
- **Define the prediction target** clearly, using either power output or capacity factor. Use only information available at the time the forecast would be issued. Future observed weather must not be used as an input to a historical forecast.

3. Model development

- Develop and compare:
 - Multiple linear regression.
 - A shallow neural network with one hidden layer.
 - A deep neural network with two or more hidden layers.
- Select and justify your **input variables** and **how far back in time they extend** (e.g. wind speed, wind direction and power output over the previous 1, 6 or 24 hours). Train and compare models with different combinations of inputs and history lengths to identify which perform best on the validation data. Use only information available when issuing the forecast.
- **Pay particular attention to input selection:** which variables you include and how much history you use can have a greater effect on forecast accuracy than hyperparameter tuning.
- First, compare different input combinations and history lengths using validation data, then refine the hyperparameters of promising models.



Wind Power Forecasting Using Machine Learning: A Swedish Case Study

Detailed Project Guidelines

4. Performance evaluation

- Include a **persistence forecast as a baseline**, assuming that future power output equals the latest known value.
- **Divide the data chronologically** into training, validation and test periods. Use rolling or expanding-window validation during model development, and reserve the final test period for evaluation after model selection.
- Fit preprocessing steps using training data only. Evaluate all models on the same test timestamps and forecast horizons.
- Report mean absolute error (MAE) and root mean squared error (RMSE), with units or a clearly stated normalisation.
- **Compare each model** with the persistence baseline and against each other using the same test data, forecast horizon, and error metrics.
- Use plots to examine predicted and reference output, forecast errors and performance under different operating conditions, including periods of rapidly changing generation.

5. Engineering interpretation and limitations

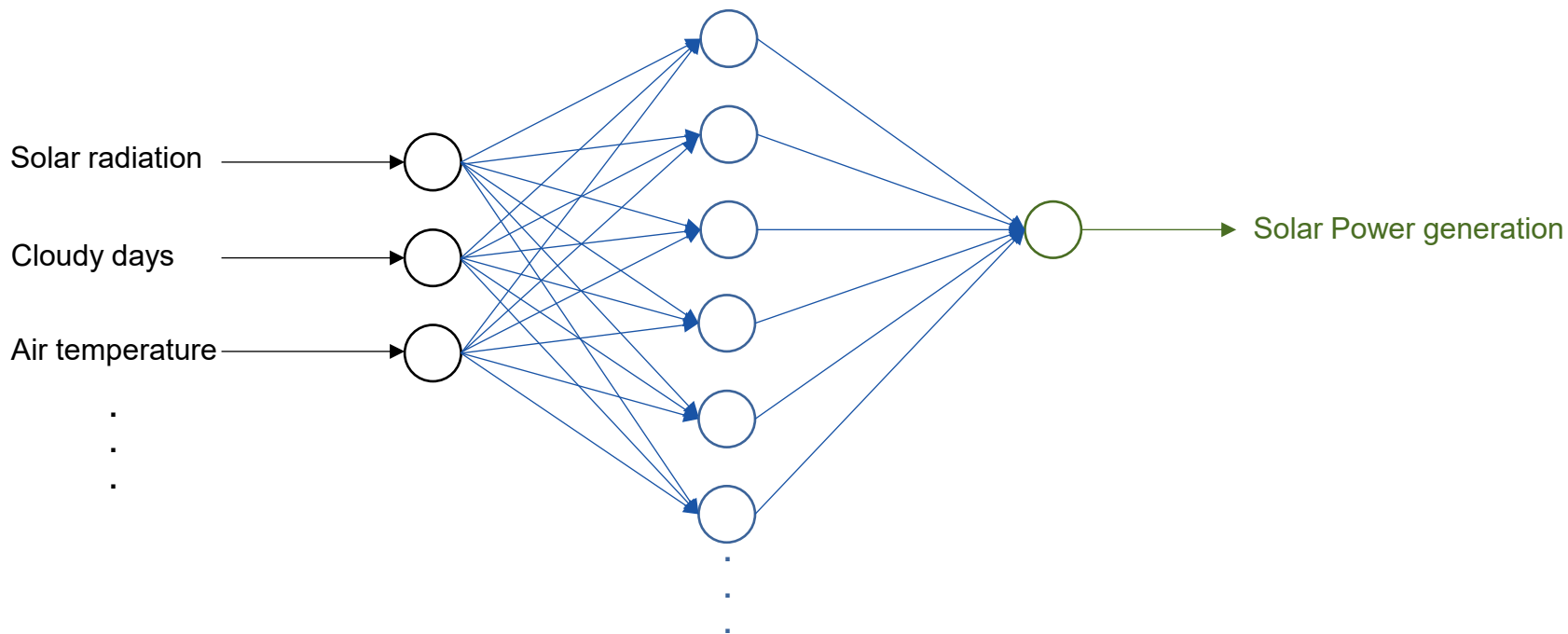
- **Discuss:**
 - Whether additional model complexity produces a meaningful improvement.
 - How forecast accuracy changes with horizon and operating conditions.
 - The consequences of overprediction and underprediction for energy-system operation.
 - How simulated data, data quality and site-specific conditions affect your conclusions.
 - What further evidence would be needed before applying the model to a real wind farm.
- **Support your conclusions** with results and acknowledge cases where a simpler model performs equally well or better.

Forecasting Solar Power Generation Using Deep Learning Techniques: A Case Study from Sweden

Farzin Golzar
fargo@kth.se



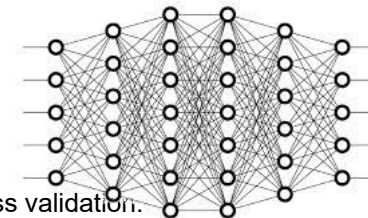
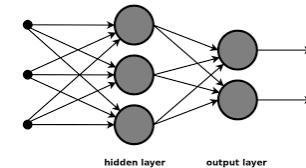
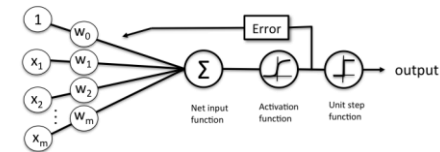
Title: Forecasting solar Power Generation Using Deep Learning Techniques: A Case Study from Sweden



Title: Forecasting Solar Power Generation Using Deep Learning Techniques: A Case Study from Sweden (Luleå, Sundsval, Stockholm, Malmo)

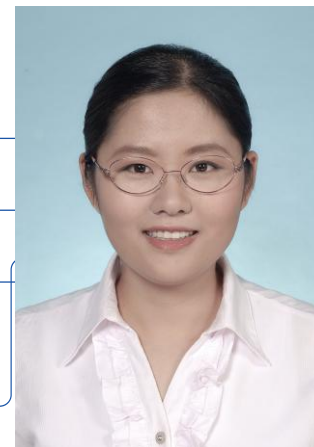
Process:

- 1) Review at least 5 relevant scientific papers to answer:
 - Which machine learning algorithms they used
 - Which input features they used in their machine learning algorithms
- 2) Based on literature review, decide which input features you want to consider.
- 3) Use renewables.ninja and smhi.se to collect historical data
- 4) Clean and prepare the data
- 5) Forecast Solar Power Generation using:
 - Logistic Regression
 - Shallow Neural Networks (1 hidden layer)
 - Deep Neural Networks (more than 1 hidden layer)
- 5) For each method, hyper tune the parameters to find the best performing parameters, and cross validation.
- 6) Compare 3 different algorithms and report the accuracy of each one
- 7) How would you address concerns related to the potential biases or inaccuracies in AI-based energy forecasts, especially when they influence critical decisions in sustainable energy systems?

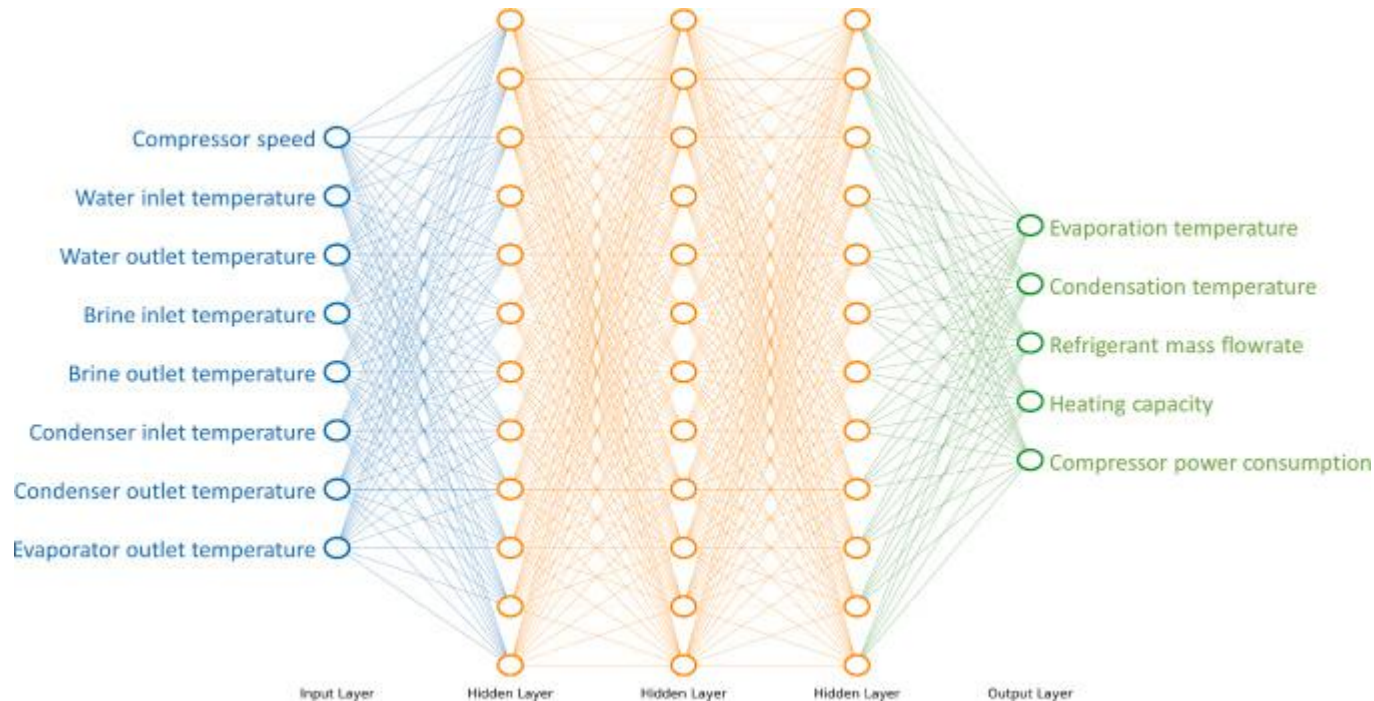


Predicting Energy Consumption in Heat Pumps

Yang Song
yson@kth.se



Predicting Energy Consumption in Heat Pumps





Steps

- 1) **Step1-** Do a detailed literature review, to assess the work done, methods used, and results obtained.
 - 2) **Step2-** Based on the initial study and review conducted, conduct a feature selection process. You need to explain how and why you choose the input features.
 - 3) **Step 3-** Cleaning the data for outliers and dealing with missing data. Explain clearly what type of data preprocessing techniques were used if any and why.
 - 4) **Step 4-** Forecast energy consumption:
 - Logistic Regression
 - Shallow Neural Networks (1 hidden layer)
 - Deep Neural Networks (more than 1 hidden layer)
 - 5) **Step 5-** For each method, hyper tune the parameters to find the best performing parameters, and cross validation.
 - 6) **Step 6-** Compare different methods and report the best performing model
 - 7) **Step 7-** Compare your results with existing literature and discuss
-

Occupancy Detection in Residential Buildings using Machine Learning

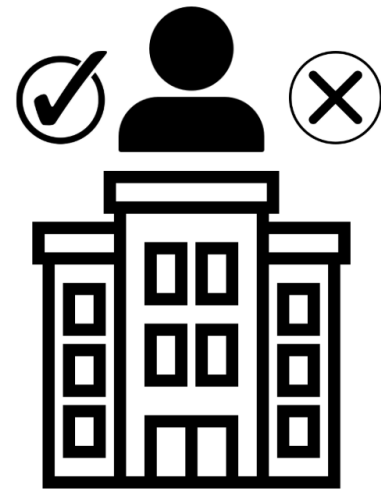
Supriya Mini Soman
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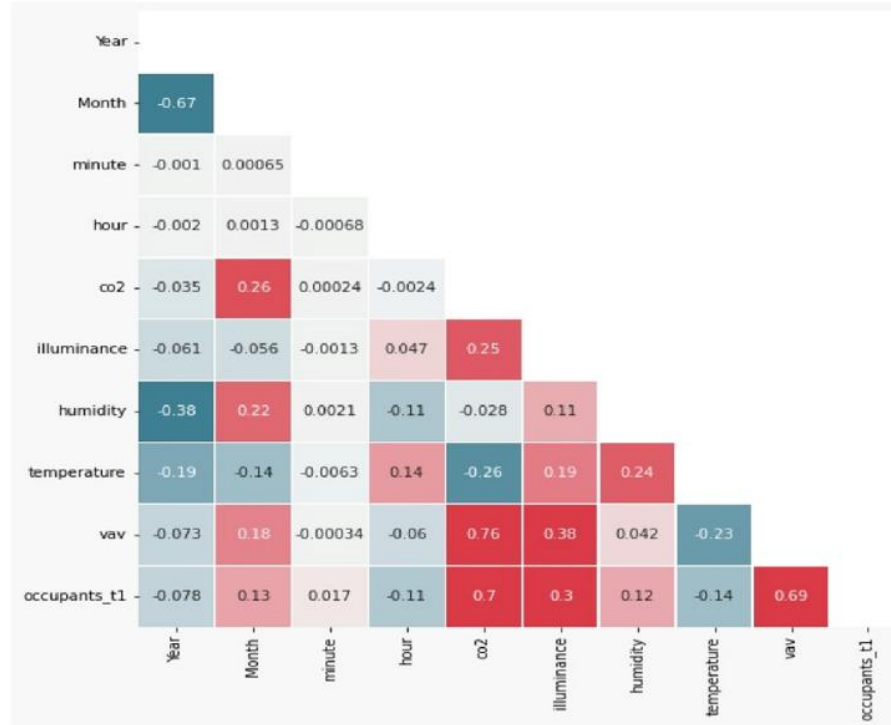


Why occupancy detection?

- Strong impact energy efficiency increment measures
- Occupancy-based recommendations from the digital twin
- Benefit to building owner
 - Cost, energy consumption and emission reduction
- Benefit to resident
 - Occupant comfort enhancement
- Highly dynamic

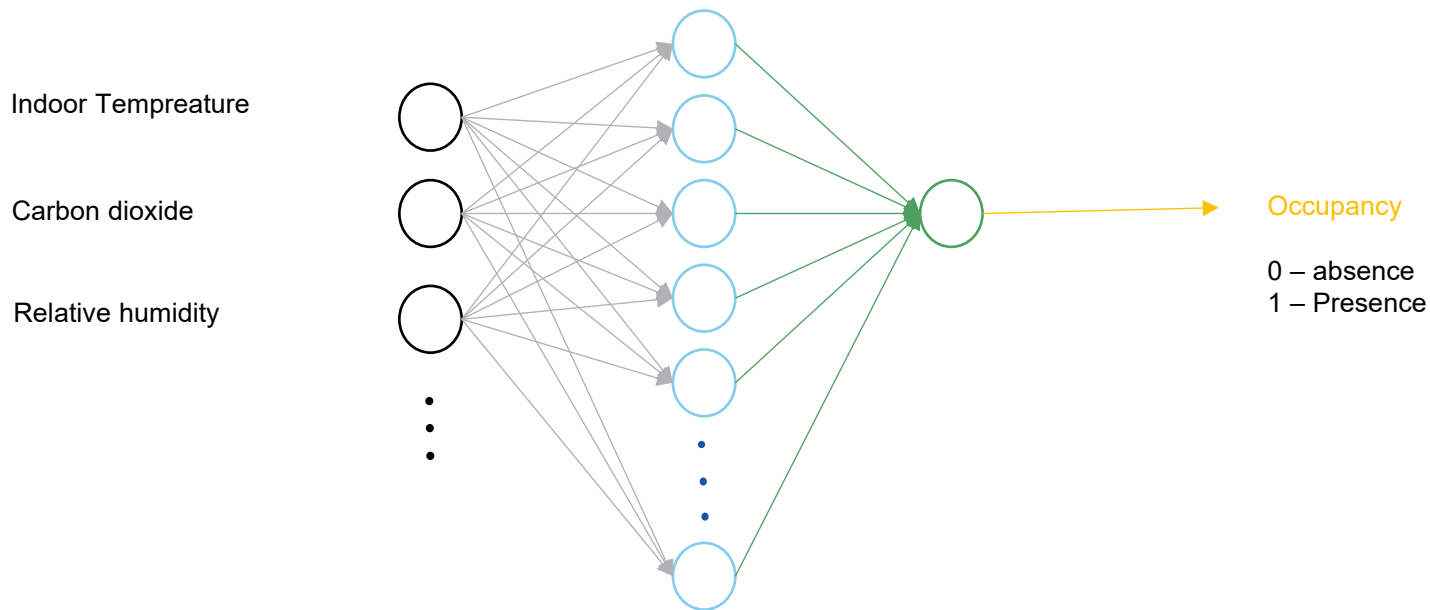


Correlation





Machine learning for occupancy detection





Steps

Each group is assigned an apartment of the KTH Live-In Lab with dataset containing environmental sensor data

1. Step 1: Literature review to understand why and how occupancy detection is done and what machine learning techniques are used in the state-of-the-art
 2. Step 2: Select the most important variables needed for training the model based on the initial research from the available dataset
 3. Step 3: Data pre-processing – analyze the data and remove any discrepancies such as missing data. Explain every discrepancy and solution to remove it
 4. Step 4: Data resampling – make the time resolution of the data uniform. Find optimal time resolution
 5. Step 5: Model training – Use one sensor data or combination of sensors to create the training set. Use:
 - i. Logistic regression
 - ii. Shallow neural network
 - iii. Deep neural network
 6. Step 6: For each model, identify and tune hyperparameters
 7. Step 7: Implement cross validation, compare different methods and report the best performing model
 8. Step 8: Discuss real-life applications and compare with existing literature results.
-



Planning

Date	Activity
Friday 18 Sep 2026 15:00 - 17:00	Projects explanation, group forming, get to know teammates and supervisors
Thursday 1 Oct 2026 08:00 - 10:00	Project progress I
Tuesday 27 Oct 2026 13:00 - 15:00	Project progress II
Wednesday 18 Nov 2026 08:00 - 12:00	Final presentation
Week 49	Reports (the template will be provided.)



Thanks you!